

CLASS TEST-1
MML

Time: 30 minutes

Max. Points: 10

Here, we follow the same notations used in class.

(1) **(4 Points)** Prove or disprove.

(i) $\{(x, y) \mid x^2 + y^2 = 0, \ x, y \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.

(ii) $\{(x, y) \mid x^2 + y^2 = 0, \ x, y \in \mathbb{C}\}$ is a subspace of $\mathbb{C}^2$.

(2) **(4 Points)** Let v_1, v_2, v_3 be linear independent in V . For what values of k are the vectors $v_2 - v_1, kv_3 - v_2, v_1 - v_3$ linearly independent?

(3) **(4 Points)** Let

$$x^6 + x^5 + x^3 + 1 = \alpha_6(x-2026)^6 + \alpha_5(x-2026)^5 + \alpha_4(x-2026)^4 + \alpha_3(x-2026)^3 + \alpha_2(x-2026)^2 + \alpha_1(x-2026) + \alpha_0.$$

Find α_1 and α_6 .

(4) **(2 Points)** Can you get a linear map from $T: \mathbb{R} \rightarrow \mathbb{R}$ such that $T(2) = 4$ and $T(5) = 25$?